

Lecture notes for March 3 to 10 (Analytic Geometry)

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1 Linear algebra and Analytic geometry

1.1 Inner product

Let V be a real vector space, an *inner product* on V is a map

$$g : V \times V \rightarrow \mathbb{R}$$

such that

- Bilinear: $g(\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2, \mathbf{v}) = \lambda_1 g(\mathbf{u}_1, \mathbf{v}) + \lambda_2 g(\mathbf{u}_2, \mathbf{v})$ for any $\lambda_1, \lambda_2 \in \mathbb{R}$ and $\mathbf{u}_1, \mathbf{u}_2, \mathbf{v} \in V$;
- Symmetric: $g(\mathbf{u}, \mathbf{v}) = g(\mathbf{v}, \mathbf{u})$ for any $\mathbf{u}, \mathbf{v} \in V$;
- Positive definite: $g(\mathbf{u}, \mathbf{u}) \geq 0$, and “=” is achieved only if $\mathbf{u} = \mathbf{0}$.

An inner product is also denoted by $\langle \cdot, \cdot \rangle_g$ or simply “ $\langle \cdot, \cdot \rangle$ ”, “ $\cdot$ ” if no confusion is arouse.

1.1.1 Normal form

For finitely dimensional space, we have the following classification results:

Theorem 1.1 (Classification of symmetric bilinear form). *If $\dim_{\mathbb{R}} V = n$, then for any symmetric bilinear form g on V , there exists basis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ such that*

$$[g(\mathbf{v}_i, \mathbf{v}_j)] = \text{diag} \left(\underbrace{1, \dots, 1}_{r_+}, \underbrace{-1, \dots, -1}_{r_-}, 0, \dots, 0 \right).$$

Here

$$r_+ = \max \{ \dim W | g|_W > 0 \text{ for a subspace } W \}$$

and

$$r_- = \{ \dim W | g|_W < 0 \text{ for a subspace } W \}$$

are called the positive inertia index and negative inertia index respectively.

Theorem 1.2 (Classification of inner product). *If $\dim_{\mathbb{R}} V = n$, then for any inner product g on V ,*

1. *there exists a basis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ such that*

$$(g(\mathbf{v}_i, \mathbf{v}_j)) = \text{diag}(1, \dots, 1),$$

i.e. orthonormal basis with respect to g exists;

2. *any basis $\{\mathbf{v}'_1, \dots, \mathbf{v}'_n\}$ is orthonormal with respect to g iff*

$$(\mathbf{v}'_1, \dots, \mathbf{v}'_n) = (\mathbf{v}_1, \dots, \mathbf{v}_n) P$$

for some $P \in O(n) = \{A \in M_{n \times n}(\mathbb{R}) \mid AA^T = I\}$, called the set of orthogonal matrix/orthogonal group.

1.1.2 In Coordinates

Let V be a real vector space, an ordered set of maximal linearly independent subset $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is called an ordered basis. Any fixed choice of ordered basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ determines the following one-to-one correspondences:

- 1.

$$\begin{aligned} V &\rightarrow \mathbb{R}^n, \\ \mathbf{v} &\mapsto \mathbf{x} \end{aligned}$$

where $\mathbf{x}$ is the coordinate column vector of $\mathbf{v}$;

- 2.

$$\begin{aligned} \{\text{linear map on } V\} &\rightarrow M_{n \times n}(\mathbb{R}), \\ \mathcal{A} &\mapsto A, \end{aligned}$$

where A is the matrix representing the linear map $\mathcal{A}$;

- 3.

$$\begin{aligned} \{\text{bilinear form on } V\} &\rightarrow M_{n \times n}(\mathbb{R}), \\ g &\mapsto (g(\mathbf{v}_i, \mathbf{v}_j)), \end{aligned}$$

- 4.

$$\{\text{symmetric bilinear form on } V\} \rightarrow \text{Sym}_{n \times n}(\mathbb{R}) = \{A \in M_{n \times n}(\mathbb{R}) \mid A^T = A\},$$

where the correspondence as above;

- 5.

$$\{\text{inner product on } V\} \rightarrow \text{Sym}_{n \times n}^+(\mathbb{R}) = \{A \in \text{Sym}_{n \times n}(\mathbb{R}) \mid A > 0\},$$

where the correspondence as above.

Under these correspondence, Theorem 1.2 and Theorem 1.1 appear in the more familiar shape

Theorem 1.3. *For any $A \in \text{Sym}_{n \times n}(\mathbb{R})$, there exists*

$$P \in \text{GL}(n, \mathbb{R}) = \{B \in M_{n \times n}(\mathbb{R}) \mid \det B \neq 0\},$$

i.e. the set of invertible matrices/General linear group, such that

$$PAP^T = \text{diag}(1, \dots, 1, -1, \dots, -1, 0, \dots, 0).$$

Theorem 1.4. *For any $A \in \text{Sym}_{n \times n}^+(\mathbb{R})$, there exists $P \in \text{GL}(n, \mathbb{R})$ such that*

$$PAP^T = \text{diag}(1, \dots, 1).$$

The proof of these theorems is based on the well-known *Spectral Theorem*:

Theorem 1.5 (Spectral Theorem/Normal form under orthogonal congruence/Normal form under orthogonal similarity/Orthogonal diagonalization). *For any $A \in \text{Sym}_{n \times n}(\mathbb{R})$, there exists $P \in O(n)$ and $\lambda_1, \dots, \lambda_n \in \mathbb{R}$ such that*

$$PAP^T = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Moreover, $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A .

1.1.3 Geometric notions induced from inner product

The following geometric notions can be induced from an inner product g on V :

- Length of a vector:

$$|\mathbf{v}|_g := \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle_g}.$$

- Angle: for $\mathbf{x}, \mathbf{y} \neq \mathbf{0}$, the angle between $\mathbf{x}, \mathbf{y}$ is defined as

$$\theta(\mathbf{x}, \mathbf{y}) = \cos^{-1} \frac{\langle \mathbf{x}, \mathbf{y} \rangle_g}{|\mathbf{x}| |\mathbf{y}|}$$

where $\cos : [0, \pi] \rightarrow [-1, 1]$. If $\mathbf{x}$ or $\mathbf{y}$ is zero, the angle is not defined. If $\theta(\mathbf{x}, \mathbf{y}) = \frac{\pi}{2}$, or equivalently $\langle \mathbf{x}, \mathbf{y} \rangle_g = 0$, we say $\mathbf{x}$ and $\mathbf{y}$ are perpendicular with respect to g , or $\mathbf{x}$ is perpendicular to $\mathbf{y}$ with respect to g , denoted by $\mathbf{x} \perp_g \mathbf{y}$.

- Orthogonal projections: let $\mathbf{b} \in V$, then

$$\begin{aligned} \text{pr}_{\mathbf{b}} : V &\rightarrow \langle \mathbf{b} \rangle = \text{Span}_{\mathbb{R}} \{\mathbf{b}\} \\ \mathbf{a} &\mapsto \left\langle \mathbf{a}, \frac{\mathbf{b}}{|\mathbf{b}|_g} \right\rangle_g \frac{\mathbf{b}}{|\mathbf{b}|_g} \end{aligned}$$

is called the “*projection onto the direction of $\mathbf{b}$* ”. We have the orthogonal decomposition

$$V = \langle \mathbf{b} \rangle \oplus \langle \mathbf{b} \rangle^{\perp_g}$$

and the “*projection onto the subspace perpendicular to $\mathbf{b}$* ”:

$$\begin{aligned} \text{pr}_{\langle \mathbf{b} \rangle^{\perp_g}} : V &\rightarrow \langle \mathbf{b} \rangle^{\perp_g}, \\ a &\mapsto \mathbf{a} - \text{pr}_{\langle \mathbf{b} \rangle} \mathbf{a}. \end{aligned}$$

- Volume of k -dimensional parallelepiped is defined inductively. Let

$$P_{\mathbf{v}_1, \dots, \mathbf{v}_k} = \left\{ \sum_{i=1}^k t_i \mathbf{v}_i \mid 0 \leq t_1, \dots, t_k \leq 1 \right\}$$

be the parallelepiped spanned by $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$, then

$$\text{Vol}_g(P_{\mathbf{v}_1, \dots, \mathbf{v}_k}) := \text{Vol}_g(P_{\mathbf{v}_1, \dots, \mathbf{v}_{k-1}}) \cdot \left| \text{pr}_{\langle \mathbf{v}_1, \dots, \mathbf{v}_{k-1} \rangle^{\perp_g}} \mathbf{v}_k \right|_g.$$

- Special case: 2-dimensional parallelepiped is called *parallelogram*, and volume is called *area* in this case, denoted by Area_g :

$$\text{Area}_g(P_{\mathbf{a}, \mathbf{b}}) = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle_g \langle \mathbf{b}, \mathbf{b} \rangle_g - \langle \mathbf{a}, \mathbf{b} \rangle_g^2}$$

- In general, we have the theorem

Theorem 1.6.

$$\text{Vol}_g(P_{\mathbf{v}_1, \dots, \mathbf{v}_k}) = \sqrt{\det(\langle \mathbf{v}_i, \mathbf{v}_j \rangle_g)}.$$

1.2 Orientation

We have heard the terminology “clockwise, anti-clockwise” when we refer to phenomenon in two-dimensional space, and “Left-Hand Rule, Right-Hand Rule.....” when we refer to phenomenon in three-dimensional space. These have relation with the abstract notion of *orientation* (定向).

Definition 1.7 (Equivalence relation). A subset $S \subset X \times X$ is called a *binary relation* on X . A binary relation S on X is called an *equivalence relation* on X if

- (Symmetry): $(x, y) \in S$ implies $(y, x) \in S$;
- (Transitivity): $(x, y), (y, z) \in S$ implies $(x, z) \in S$;
- (Reflexivity): $(x, x) \in S$ for any $x \in X$.

An equivalence relation S is usually denoted by “ $\sim_S$ ”, or simply “ $\sim$ ” if no confusion arise.

Example 1.8 (Equivalence class). Let “ $\sim$ ” be an equivalence relation on X , then the set

$$\{y \in X | x \sim y\},$$

denoted by $[x]_{\sim}$ is called the equivalence class of x under the equivalence relation $\sim$. Any $y \in [x]_{\sim}$ is called a *representative* of the equivalence class. The set

$$X/\sim := \{[x]_{\sim} | x \in X\} \subset \mathcal{P}(X)$$

is called the *set of equivalence classes* under the equivalence relation $\sim$, or the *quotient set*.

Example 1.9.

1. The set $\{(x, y) \in \mathbb{R} \times \mathbb{R} | x > y\}$ is a binary relation on $\mathbb{R}$, which we usually denote by “ $>$ ” for simplicity. It is not an equivalence relation.
2. The diagonal subset $S = \{(x, x) | x \in X\} \subset X \times X$ is always an equivalence relation on X , which we usually denote by “ $=$ ”.
3. (Congruence) Let $n \in \mathbb{N}$, the set $\{(x, y) \in \mathbb{Z} \times \mathbb{Z} | n | (x - y)\}$ is an equivalence relation, denoted by $\equiv_n$, is called the “congruence relation mod n ”. Each equivalence class $[k]_{\equiv_n}$ is called a *mod n congruence class*, and the set of mod n congruence classes is denoted by

$$\mathbb{Z}_n := \mathbb{Z} / \equiv_n.$$

4. (Similarity) We can define $\sim$ on $M_{n \times n}(\mathbb{R})$ by

$$“A \sim B” \text{ iff } “A = PBP^{-1} \text{ for some } P \text{ invertible}”.$$

This is an equivalence relation on $M_{n \times n}(\mathbb{R})$, called *similarity*.

Definition 1.10 (Orientation, Oriented vector space). Let $\mathcal{B}_V = \{\text{ordered basis of } V\}$ and the equivalence relation “ $\sim$ ” defined as:

$$(\mathbf{v}'_1, \dots, \mathbf{v}'_n) \sim (\mathbf{v}_1, \dots, \mathbf{v}_n)$$

iff

$$(\mathbf{v}'_1, \dots, \mathbf{v}'_n) = (\mathbf{v}_1, \dots, \mathbf{v}_n)P \text{ for some } P \text{ with } \det P > 0.$$

The quotient set $\mathcal{B}_V / \sim$ consists of exactly two elements, each of which is called an *orientation*. Each real vector space V , together with a choice of orientation “ o ”, is called an *oriented vector space*.

1.3 Cross product

Recall that $\mathbb{R}^n := \{(x_1, \dots, x_n)^T | x_1, \dots, x_n \in \mathbb{R}\}$ equipped with the standard inner product

$$\langle \mathbf{x}, \mathbf{y} \rangle := \sum_{i=1}^n x_i y_i$$

for $\mathbf{x} = (x_1, \dots, x_n)^T, \mathbf{y} = (y_1, \dots, y_n)^T$, is called the *n-dimensional Euclidean space*. The ordered set

$$(\mathbf{e}_1, \dots, \mathbf{e}_n)$$

where $\mathbf{e}_i = (0, \dots, 1, \dots, 0)^T$, is called the *standard orthonormal basis*.

In the special case $n = 2$, $\mathbf{e}_1, \mathbf{e}_2$ is usually denoted by $\mathbf{i}$ and $\mathbf{j}$, and in case $n = 3$, $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ are usually denoted by $\mathbf{i}, \mathbf{j}, \mathbf{k}$.

Definition 1.11. Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^3$, the vector

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} = \begin{vmatrix} x_2 & x_3 \\ y_2 & y_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} x_1 & x_3 \\ y_1 & y_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} \mathbf{k}$$

is called the *cross product* of $\mathbf{x}$ and $\mathbf{y}$.

It has the following properties:

1. $\mathbf{x} \times \mathbf{y} = -\mathbf{y} \times \mathbf{x}$;
2. $\langle \mathbf{x} \times \mathbf{y}, \mathbf{x} \rangle = \langle \mathbf{x} \times \mathbf{y}, \mathbf{y} \rangle = 0$, and $(\mathbf{x}, \mathbf{y}, \mathbf{x} \times \mathbf{y})$ determines the same orientation as $(\mathbf{i}, \mathbf{j}, \mathbf{k})$;
3. $|\mathbf{x} \times \mathbf{y}| = \text{Area}(P_{\mathbf{x}, \mathbf{y}})$. This follows from

$$\begin{aligned} & (x_1^2 + x_2^2 + x_3^2)(y_1^2 + y_2^2 + y_3^2) - (x_1 y_1 + x_2 y_2 + x_3 y_3)^2 \\ &= (x_2 y_3 - x_3 y_2)^2 + (x_3 y_1 - x_1 y_3)^2 + (x_1 y_2 - x_2 y_1)^2. \end{aligned}$$

4. It is bilinear, i.e. $(\lambda \mathbf{x} + \mu \mathbf{y}) \times \mathbf{z} = \lambda \mathbf{x} \times \mathbf{z} + \mu \mathbf{y} \times \mathbf{z}$.

Geometrically, $\mathbf{x} \times \mathbf{y}$ represents the vector that whose length is equal to the area of the parallelepiped spanned by $\mathbf{x}$ and $\mathbf{y}$, and whose direction is determined by the Right-Hand Rule.

Theorem 1.12. In $\mathbb{R}^3$, the cross product is uniquely determined by the four properties.

Definition 1.13 (Mixed product in $\mathbb{R}^3$). Let $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$,

$$\langle \mathbf{a}, \mathbf{b} \times \mathbf{c} \rangle = \langle \mathbf{b}, \mathbf{c} \times \mathbf{a} \rangle = \langle \mathbf{c}, \mathbf{a} \times \mathbf{b} \rangle = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

is called the *mixed product* of the vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$.

It is positively iff $(\mathbf{a}, \mathbf{b}, \mathbf{c})$ determines the same orientation as $(\mathbf{i}, \mathbf{j}, \mathbf{k})$, i.e. $(\mathbf{a}, \mathbf{b}, \mathbf{c})$ is a Right-Hand system. From the geometric meaning, $|\langle \mathbf{a}, \mathbf{b} \times \mathbf{c} \rangle| = \text{Vol}(P_{\mathbf{a}, \mathbf{b}, \mathbf{c}})$. So the mixed product also measures the “signed volume” of a parallelepiped.

In $\mathbf{R}^n$, if we write $U_{n \times k}$ as the coordinate matrix of the vectors $(\mathbf{v}_1, \dots, \mathbf{v}_k)$, i.e.

$$(\mathbf{v}_1, \dots, \mathbf{v}_k) = (\mathbf{e}_1, \dots, \mathbf{e}_n) U,$$

then

$$\begin{aligned} \mathbf{v}_i &= \sum_{k=1}^n U_{ki} \mathbf{e}_k, \\ \langle \mathbf{v}_i, \mathbf{v}_j \rangle &= \sum_{k,l=1}^n U_{ki} U_{lj} \delta_{kl} = (U^T U)_{ij}. \end{aligned}$$

As a consequence, we have the

Theorem 1.14 (Generalized Pythagorean Theorem).

$$\text{Vol}_g(P_{\mathbf{v}_1, \dots, \mathbf{v}_k}) = \sqrt{\sum_{1 \leq i_1 < \dots < i_k \leq n} U \begin{pmatrix} i_1 i_2 \dots i_k \\ 12 \dots k \end{pmatrix}^2}$$

where

$$U \begin{pmatrix} i_1 i_2 \dots i_k \\ 12 \dots k \end{pmatrix} = \det \begin{pmatrix} U_{i_1 1} & \dots & U_{i_1 k} \\ \vdots & \ddots & \vdots \\ U_{i_k 1} & \dots & U_{i_k k} \end{pmatrix}$$

represents the volume of the projected parallelepiped of $P_{\mathbf{v}_1, \dots, \mathbf{v}_k}$ onto the subspace $\langle \mathbf{v}_{i_1}, \dots, \mathbf{v}_{i_k} \rangle$.

1.4 Lines and Planes in $\mathbb{R}^3$

The two space $\mathbb{R}^2$ and $\mathbb{R}^3$ are the two space where classical Euclidean geometry takes place on. The familiar notions in Euclidean geometry are

- Point: $(x_1, x_2, x_3)^T$;
- Line: 1-dimensional affine subspace, i.e. shift of 1-dimensional subspace, is called a line in $\mathbb{R}^3$. There are two ways of representing a line in $\mathbb{R}^3$:
 - (Point-Direction form: Explicit equation) This gives a parameter equation :

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}, t \in \mathbb{R}.$$

In terms coordinates, this is

$$x = x_0 + lt, y = y_0 + mt, z = z_0 + nt.$$

- (Plane-intersection: Implicit equation) Take the *Point-Direction form* the line, $\mathbf{r}$ lies on the line iff

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v},$$

we get the equation

$$(\mathbf{r} - \mathbf{r}_0) \times \mathbf{v} = \mathbf{0}.$$

In terms of coordinates, it reads

$$\begin{bmatrix} n & -m \\ -n & l \\ m & -l \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} ny_0 - mz_0 \\ -nx_0 + lz_0 \\ mx_0 - ly_0 \end{bmatrix}.$$

This is of the form

$$\begin{cases} a_1x + b_1y + c_1z + d_1 = 0, \\ a_2x + b_2y + c_2z + d_2 = 0, \end{cases}$$

where the coefficient matrix has rank 2, which is called the *general equation* for line. The line is perpendicular to the normal vector $\mathbf{n}_1 = (a_1, b_1, c_1)^T$ of the plane Π_1 , and the normal vector $\mathbf{n}_2 = (a_2, b_2, c_2)^T$ of the plane Π_2 . Thus, the direction of the line is given by $\mathbf{n}_1 \times \mathbf{n}_2$. Together with a particular point P_0 on the line, we get the *Point-Direction form* of the line.

- Plane: 2-dimensional affine subspace, i.e. shift of 2-dimensional subspace is called a plane in $\mathbb{R}^3$.

- (Point-Direction form: Explicit equation) Take $\mathbf{r}_0$ on the plane, and any two points Q, R in the plane such that $\mathbf{u} = \overrightarrow{PQ}$ and $\mathbf{v} = \overrightarrow{PR}$ are independent, then the plane has a parameter equation

$$\mathbf{r} = \mathbf{r}_0 + s\mathbf{u} + t\mathbf{v}, s, t \in \mathbb{R}.$$

In terms of coordinate, this is

$$x = x_0 + su_1 + tv_1, y = y_0 + su_2 + tv_2, z = z_0 + su_3 + tv_3.$$

- (Point-Normal form: Implicit equation) Take a point P_0 on the plane, and let $\mathbf{n} = (a, b, c)^T$ be its normal vector, the equation is

$$\langle \mathbf{n}, \mathbf{r} - \mathbf{r}_0 \rangle = 0.$$

In terms of coordinate, this is

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

or

$$ax + by + cz + d = 0,$$

which is called a *general equation* for the plane (平面的一般方程).

1.4.1 Relations between two planes

Suppose we have two planes:

$$\begin{aligned}\pi_1 : a_1x + b_1y + c_1z + d_1 &= 0 \\ \pi_2 : a_2x + b_2y + c_2z + d_2 &= 0.\end{aligned}$$

The normal vectors are $\mathbf{n}_1 = (a_1, b_1, c_1)^T$ and $\mathbf{n}_2 = (a_2, b_2, c_2)^T$ respectively. About the relation between the positions of the two planes there are two cases

- (1) Parallel: $\mathbf{n}_1 \parallel \mathbf{n}_2$, i.e. there exists $\lambda \neq 0$ such that $\mathbf{n}_1 = \lambda \mathbf{n}_2$. If $d_1 = \lambda d_2$, then π_1 coincides with π_2 ;
- (2) Intersecting: we define the angle between π_1 and π_2 to be the angle between $\mathbf{n}_1$ and $\mathbf{n}_2$, i.e.

$$\theta = \arccos \frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{|\mathbf{n}_1| |\mathbf{n}_2|}.$$

In particular, if $\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$, then $\pi_1 \perp \pi_2$.

Example 1.15. Find the plane through z -axis and perpendicular to $x - 2y + 5 = 0$.

Solution. Since it passes through z -axis, we can write its equation as

$$Ax + By = 0. \tag{1.1}$$

Perpendicular to $x - 2y + 5 = 0$ means that

$$(A, B, 0) \cdot (1, -2, 0) = 0.$$

So we can take $A = 2, B = 1$, and the equation is $2x + y = 0$.

Example 1.16. Find the plane through the points $P_1(8, -3, 1), P_2(4, 7, 2)$ and perpendicular to $\pi : 3x + 5y - 7z + 21 = 0$.

Solution. The unknown normal vectors $\mathbf{n}$ must be perpendicular to $\overrightarrow{P_1P_2}$ and $\mathbf{n}_1 = (3, 5, -7)$, therefore we can take

$$\mathbf{n} = \mathbf{n}_1 \times \overrightarrow{P_1P_2} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 5 & -7 \\ -4 & 10 & 1 \end{vmatrix} = 75\mathbf{i} + 25\mathbf{j} + 50\mathbf{k}.$$

Therefore, the plane in question should be

$$3(x - 8) + (y + 3) + 2(z - 1) = 3x + y + 2z - 23 = 0.$$

Example 1.17. Find the angle between the two planes $x - y + z = 0$ and $x + y - z = 0$.

1.4.2 The distance from point to plane

Given the plane $\pi : ax + by + cz + d = 0$ and a point $P_0(x_0, y_0, z_0)$ outside π , what is then the distance from P_0 to π ?

Take arbitrary $P(x, y, z)$ on π , then the distance is

$$\left| \overrightarrow{PP_0} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right| = \left| \frac{a(x_0 - x) + b(y_0 - y) + c(z_0 - z)}{\sqrt{a^2 + b^2 + c^2}} \right| = \left| \frac{ax_0 + by_0 + cz_0 + d}{\sqrt{a^2 + b^2 + c^2}} \right|$$

The sign of the quantity $ax_0 + by_0 + cz_0 + d$ represents which side P_0 is relative to the plane π . If we define the part which $\mathbf{n}$ is pointed to as upper half space, then P_0 is in the upper half iff $ax_0 + by_0 + cz_0 + d > 0$.

1.4.3 Relation between two lines

Given two lines

$$\begin{aligned} L_1 : \mathbf{r} &= \mathbf{r}_1 + t\mathbf{v}_1 \\ L_2 : \mathbf{r} &= \mathbf{r}_2 + t\mathbf{v}_2, \end{aligned} \tag{1.2}$$

they are coplanar iff the three vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{r}_2 - \mathbf{r}_1\}$ are coplanar, i.e. $(\mathbf{r}_2 - \mathbf{r}_1) \cdot \mathbf{v}_1 \times \mathbf{v}_2 = 0$, or in terms of coordinates

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0.$$

The angle between the two lines is the angle between the two directions $\mathbf{v}_1, \mathbf{v}_2$. Thus, two lines are perpendicular iff $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$.

1.4.4 The distance between point and line

Let $L : \mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$, where $\mathbf{r}_0 = \overrightarrow{OP_0}$ for $P_0(x_0, y_0, z_0)$ and $P_1(x_1, y_1, z_1)$ be another point. The distance from P_1 to L is

$$d = |\overrightarrow{P_0P_1}| \sin \theta = \frac{|\mathbf{v} \times \overrightarrow{P_0P_1}|}{|\mathbf{v}|} = \frac{\dots}{\sqrt{l^2 + m^2 + n^2}}.$$

1.4.5 The relation between the position of line and plane

Given a line and a plane as follows:

$$\begin{aligned} L : \mathbf{r} &= \mathbf{r}_0 + t\mathbf{v}, \quad \mathbf{v} = (l, m, n) \\ \pi : (\mathbf{r} - \mathbf{r}_1) \cdot \mathbf{n} &= 0, \quad \mathbf{n} = (a, b, c). \end{aligned} \tag{1.3}$$

The line L projects (orthogonal) to L' on π , and the angle φ between L and L' is defined to be the angle between the line L and the plane π . By geometric interpretation:

$$\sin \varphi = |\cos \theta(\mathbf{n}, \mathbf{v})| = \frac{|\mathbf{n} \cdot \mathbf{v}|}{|\mathbf{n}||\mathbf{v}|} = \frac{|al + bm + cn|}{\sqrt{a^2 + b^2 + c^2} \sqrt{l^2 + m^2 + n^2}}.$$

- $L \perp \pi$ iff $\mathbf{v} \parallel \mathbf{n}$;
- $L \parallel \pi$ iff $\mathbf{v} \perp \mathbf{n}$;
- $L \cap \pi = \text{point}$ iff $\sin \varphi < 1$. i.e. $\mathbf{v} \cdot \mathbf{n} \neq 0$, then we can solve the intersection point $P'(x', y', z')$ by solving:

$$(\mathbf{r}' - \mathbf{r}_1) \cdot \mathbf{n} = 0, \quad \mathbf{r}' = \mathbf{r}_0 + t' \mathbf{v}.$$

$$\text{We get } t' = -\frac{(\mathbf{r}_0 - \mathbf{r}_1) \cdot \mathbf{n}}{\mathbf{v} \cdot \mathbf{n}}.$$

1.5 Conic section/curve and Quadric surface

We are going to use orthogonal transformations, or more generally Euclidean motions, to understand simple geometric object defined in Euclidean spaces.

As a preparation, let us look at a plane defined by the general equation

$$ax + by + cz + d = 0. \quad (1.4)$$

Consider a coordinate transformation

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = P \begin{pmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{pmatrix}, \quad (1.5)$$

then the above equation is changed to

$$(a \ b \ c) P \begin{pmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{pmatrix} + d = 0.$$

If we chose $P \in O(3)$ such that

$$(a \ b \ c) P = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

then the equation becomes a very simple form

$$\tilde{z} + d = 0. \quad (1.6)$$

Such P of course exists, and actually in abundance, for instance we can take $P = (\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3)$ where

$$\mathbf{p}_1 = \frac{1}{\sqrt{a^2 + b^2 + c^2}} (a \ b \ c)^T, \quad \mathbf{p}_2 \perp \mathbf{p}_1, \quad \mathbf{p}_3 = \mathbf{p}_1 \times \mathbf{p}_2.$$

If we allow more general transformation

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = P \begin{pmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{pmatrix} + \mathbf{q}$$

with $\mathbf{q} = -\frac{d}{\sqrt{a^2+b^2+c^2}}\mathbf{p}_1$, then the equation (1.4) becomes even more simpler

$$\tilde{z} = 0. \quad (1.7)$$

Definition 1.18 (Conic section/curve in $\mathbb{R}^2$). The set $\{(x, y) \in \mathbb{R}^2 | f(x, y) = 0\}$ where $f(x, y)$ is a real degree 2 polynomial in x and y is called a *conic curve* or *conic section*.

Definition 1.19 (Quadric surface in $\mathbb{R}^3$). The set $\{(x, y, z) \in \mathbb{R}^3 | f(x, y, z) = 0\}$ where $f(x, y, z)$ is a real degree 2 polynomial in x, y, z is called a *quadric surface*.

1.5.1 General scheme

To simplify notations, we consider

$$f(x_1, \dots, x_n) = \sum_{i,j=1}^n x_i A_{ij} x_j + \sum_{i=1}^n b_i x_i + c = \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x} + c$$

where $A = (A_{ij})$ is a symmetric $n \times n$ -matrix. Consider the linear coordinate transformation

$$\mathbf{x} = P\mathbf{y}, \quad (1.8)$$

then the equation becomes

$$\mathbf{y}^T P^T A P \mathbf{y} + \mathbf{b}^T P \mathbf{y} + c = 0. \quad (1.9)$$

By the Spectral Theorem, we can find $P \in SO(n) = \{P \in O(n) | \det P = 1\}$ such that

$$P^T A P = \Lambda = \text{diag} \left(\underbrace{\lambda_1, \dots, \lambda_{r_+}}_{r_+}, \underbrace{-\lambda_{r_++1}, \dots, -\lambda_{r_++r_-}}_{r_-}, 0, \dots, 0 \right) \quad (1.10)$$

where $\lambda_1, \dots, \lambda_{r_+}$ are the strictly positive eigenvalues of A , and $-\lambda_{r_++1}, \dots, -\lambda_{r_++r_-}$ are the strictly negative eigenvalues of A , and $r_+ + r_- = \text{rank } A$. Using such P , the equation becomes

$$\sum_{i=1}^{r_+} \lambda_i y_i^2 - \sum_{i=r_++1}^{r_++r_-} \lambda_i y_i^2 + \sum_{i=1}^n \tilde{b}_i y_i + c = 0.$$

A further coordinate transformation

$$\begin{pmatrix} y_1 \\ \dots \\ y_{r_+} \\ y_{r_++1} \\ \dots \\ y_{r_++r_-} \\ y_{r+1} \\ \dots \\ y_n \end{pmatrix} = \begin{pmatrix} z_1 - \frac{\tilde{b}_1}{2\lambda_1} \\ \dots \\ y_{r_+} = z_{r_+} - \frac{\tilde{b}_{r_+}}{2\lambda_{r_+}} \\ y_{r_++1} = z_{r_++1} + \frac{\tilde{b}_{r_++1}}{2\lambda_{r_++1}} \\ \dots \\ y_{r_++r_-} = z_{r_++r_-} + \frac{\tilde{b}_{r_++r_-}}{2\lambda_{r_++r_-}} \\ y_{r+1} \\ \dots \\ y_n \end{pmatrix} \quad (1.11)$$

change the equation to

$$\sum_{i=1}^{r_+} \lambda_i z_i^2 - \sum_{j=r_++1}^r \lambda_j z_j^2 + \sum_{k=r_++1}^n \tilde{b}_k z_k + \tilde{c} = 0$$

where

$$\tilde{c} = c - \sum_{i=1}^{r_+} \frac{\tilde{b}_i^2}{4\lambda_i} + \sum_{j=r_++1}^r \frac{\tilde{b}_j^2}{4\lambda_j}.$$

Depending on if $(\tilde{b}_{r_++1} \dots \tilde{b}_n) = (0 \dots 0)$ or not, we can further simplify the *first order term* and *zeroth order term* using an Euclidean motion in the variables $z_{r_++1}, \dots, z_n$.

Summarizing the above scheme, we have

Theorem 1.20. *There exists coordinate transformation of the form $\mathbf{x} = R\mathbf{z} + \mathbf{r}$ where $R \in SO(n)$ and $\mathbf{r} \in \mathbb{R}^n$, such that the equation $f(\mathbf{x}) = 0$ becomes*

1.

$$\sum_{i=1}^{r_+} \lambda_i z_i^2 - \sum_{j=r_++1}^r \lambda_j z_j^2 = c,$$

2.

$$\sum_{i=1}^{r_+} \lambda_i z_i^2 - \sum_{j=r_++1}^r \lambda_j z_j^2 - z_{r_++1} = 0.$$

1.5.2 Classifications of quadric surface in $\mathbb{R}^3$

Specialized to $\mathbb{R}^3$, we get the classification:

1. (rank $A = 3$): The simplified equation reads

$$\lambda_1 x'^2 + \lambda_2 y'^2 + \lambda_3 z'^2 + c = 0$$

The shape depends on the signs of the coefficients: if the sign of $(\lambda_1, \lambda_2, \lambda_3, c')$ (WLOG assume $\lambda_1 > 0$)

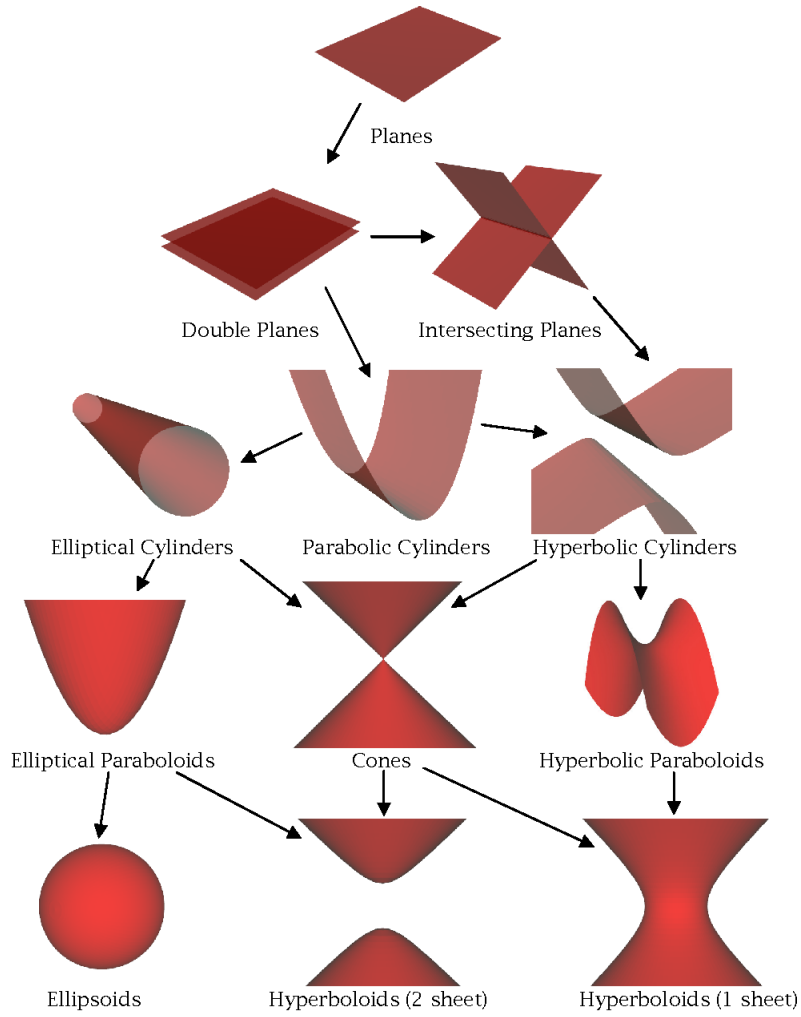


Figure 1: Quadric surfaces

- is $(+, +, +, -)$, then the shape of the quadric surface is a [ellipsoid](#), for instance $\frac{x'^2}{\mu_1^2} + \frac{y'^2}{\mu_2^2} + \frac{z'^2}{\mu_3^2} = 1$;
- is $(+, +, +, 0)$, then the set is a single point, for instance $\frac{x'^2}{\mu_1^2} + \frac{y'^2}{\mu_2^2} + \frac{z'^2}{\mu_3^2} = 0$;
- is $(+, +, -, +)$, then the shape is a [double-sheets hyperboloid](#), for instance $\frac{x'^2}{\mu_1^2} + \frac{y'^2}{\mu_2^2} - \frac{z'^2}{\mu_3^2} + 1 = 0$;
- is $(+, +, -, -)$, then the image is a [single-sheet hyperboloid](#), for instance $\frac{x'^2}{\mu_1^2} + \frac{y'^2}{\mu_2^2} - \frac{z'^2}{\mu_3^2} = 1$;

- is $(+, +, -, 0)$, then the shape is **cone**, for instance $\frac{x'^2}{\mu_1^2} + \frac{y'^2}{\mu_2^2} - \frac{z'^2}{\mu_3^2} = 0$, this is a **singular** surface with the tip the singularity;

2. (rank $A = 2$): Suppose $\lambda_3 = 0$, then the equation can be simplified to

$$\lambda_1 x'^2 + \lambda_2 y'^2 + b'_3 z' + c' = 0.$$

- If $b'_3 \neq 0$, and $\lambda_1 \lambda_2 > 0$, then the shape is a **elliptical paraboloid** surface, for instance $\frac{x'^2}{\mu_1^2} + \frac{y'^2}{\mu_2^2} + z' + 1 = 0$;
- If $b'_3 \neq 0$, and $\lambda_1 \lambda_2 < 0$, then the shape is a **hyperbolic paraboloid** surface, or saddle surface, for instance $\frac{x'^2}{\mu_1^2} - \frac{y'^2}{\mu_2^2} + z' + 1 = 0$;
- If $b'_3 = 0$, then the surface is a cylindrical surface formed by translating a conic curve in the coordinate plane $Ox'y'$ to the third direction, for instance $\frac{x'^2}{\mu_1^2} + \frac{y'^2}{\mu_2^2} + 1 = 0$ (**elliptical cylinder**), $\frac{x'^2}{\mu_1^2} - \frac{y'^2}{\mu_2^2} = 0$ (**intersecting planes**, which is **singular quadric surface** with singularities along a line) or $\frac{x'^2}{\mu_1^2} - \frac{y'^2}{\mu_2^2} = 1$ (**hyperbolic cylinder**);

3. (rank $A = 1$): Suppose $\lambda_1 \neq 0$, then the equation is simplified as

$$\lambda_1 x'^2 + b'_2 y'' + b'_3 z'' + c' = 0.$$

- b'_2 and b'_3 are not simultaneously 0, we can use a rotation in the $Oy'z'$ plane to reduce the linear terms only involving one variable, this represents a **parabolic cylinder** surface;
- If $b'_2 = b'_3 = 0$, then the surface is

$$x'^2 + \frac{c'}{\lambda_1} = 0,$$

which is either **double planes** (disjoint parallel planes) or a **plane** (with multiplicity two) or empty set.

Recall that the simplification of the equations are all achieved by a coordinate transformation of the type

$$\mathbf{x} = Q\mathbf{x}' + \mathbf{v},$$

where $Q \in SO(3)$, and $\mathbf{v} \in \mathbf{R}^3$. The geometric meaning is that under the new Cartesian coordinate system $\mathbb{F}' = [O'; \mathbf{i}', \mathbf{j}', \mathbf{k}']$ where

- O' has position vector $\mathbf{v}$, i.e. $\overrightarrow{OO'} = v_1 \mathbf{i} + v_2 \mathbf{j} + v_3 \mathbf{k}$;
- $(\mathbf{i}' \mathbf{j}' \mathbf{k}') = (\mathbf{i} \mathbf{j} \mathbf{k})Q$,

the surface $\{f(x, y, z) = 0\}$ is defined by the corresponding simplified equation

$$\tilde{f}(x', y', z') = 0.$$

Equivalently, and if we would like to visualize the change in one fixed coordinate, this means the set $C = \{\mathbf{x} | f(\mathbf{x}) = 0\}$ is obtained by applying an orthogonal transformation (represented by Q), and then a translation (represented by $\mathbf{v}$), from the set $\{\mathbf{x} | \tilde{f}(\mathbf{x}) = 0\}$.

Example 1.21. Let $\pi : x + 2y + 3z = 1$ be a plane in $\mathbf{R}^3$, find another Cartesian coordinate system $\mathbb{F}' = [O'; \mathbf{i}', \mathbf{j}', \mathbf{k}']$ such that π is the coordinate plane $x' = 0$.

Example 1.22. Let γ be the intersection of the surface $x^2 + y^2 + z^2 = 1$ and the plane $\pi : x + y + z = 0$. Find the equation for the projection curve $\tilde{\gamma}$ onto the plane Oxy , then identify the shape of γ and find an parameterization for it.

Example 1.23. Write down the equation of the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ in the Cartesian coordinate system $\mathbb{F}' = \left[(1, 1); \frac{1}{\sqrt{2}}(\mathbf{i} + \mathbf{j}), \frac{1}{\sqrt{2}}(-\mathbf{i} + \mathbf{j})\right]$.

Solution. The coordinate transformation reads as

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

therefore, the equation is changed to

$$\frac{1}{9} \left(\frac{x-y}{\sqrt{2}} + 1 \right)^2 + \frac{1}{4} \left(\frac{x+y}{\sqrt{2}} + 1 \right)^2 = 1.$$

Example 1.24. Discuss all the possible situation of the intersection of a plane and a quadric surface in $\mathbb{R}^3$.

Example 1.25. Consider the curve C formed by intersecting the parabolic cylinder $y = x^2$ and sphere $x^2 + y^2 + z^2 = 1$.

1. Find a parameterization of the curve;
2. Describe the curves on yOz, zOx, xOy planes which are the corresponding orthogonal projection of C .

1.5.3 Surface of revolution

By rotating a curve along a line in $\mathbf{R}^3$, we get a so-called *surface of revolution*. For example, let S be the surface obtained by rotating the curve $\{\frac{y^2}{a^2} + \frac{z^2}{b^2} = 1, x = 0\}$ along the z axis, then S can be represented by the equation

$$\frac{x^2 + y^2}{a^2} + \frac{z^2}{b^2} = 1.$$

In general, let $F(y, z) = 0$ be a curve γ in Oyz plane, then the surface formed by rotating γ along the z -axis is

$$F(\pm\sqrt{x^2 + y^2}, z) = 0,$$

which means that any point $P(x, y, z)$ satisfies at least one of the two equations $F(\sqrt{x^2 + y^2}, z) = 0, F(-\sqrt{x^2 + y^2}, z) = 0$. The sign $\pm$ is important especially when there are points on the curve γ with y coordinate both positive and negative. For instance, if $F(y, z) = y - z$. Then the surface of revolution is

$$\pm\sqrt{x^2 + y^2} = z.$$

If we instead only take $\sqrt{x^2 + y^2}$, we only get the upper half of the double cone.

And similarly, we can get the surface of revolution along other axis, for instance $F(y, \pm\sqrt{x^2 + z^2}) = 0$ represents rotation along the y -axis.

A point P lies on the surface formed by revolving γ along the axis L iff the planar circle centered on L and passing P intersects γ . This constraint condition provides us the following scheme of obtaining the general equation is:

- 1) For a general point $P(u, v, w)$, find the plane π passing P and perpendicular to L ;
- 2) Find $P' \in \pi \cap \gamma$ in terms of u, v, w ;
- 3) Write down the equation $d(P, L) = d(P', L)$ and simplify the equation in terms of u, v, w .

Example 1.26.

- Find the general equation for the surface formed by revolving $\gamma : y = z, x = 0$ along the z -axis.
- Find the general equation for the surface formed by revolving $\gamma : (y - 2)^2 + z^2 = 1$ along the z -axis.
- Find the general equation for the surface formed by revolving the line $\gamma : \mathbf{r}(t) = (0, 1, 0) + t(1, 0, 1)$ along z -axis.

Among the many types of quadric surfaces, some of them are ruled surfaces: all types of cylinders, cones and single-sheet hyperboloid.

Mentioned above, we can use the division of $\mathbb{R}^3$ into unions of hyperboloids and double cone to give another interesting coordinate systems to $\mathbf{R}^3$, let

$$\mathbf{R}^3 = H^+ \cup H^0 \cup H^-,$$

where $H^+ = \{(x, y, z) | x^2 + y^2 - z^2 > 0\}$, $H^0 = \{(x, y, z) | x^2 + y^2 + z^2 = 0\}$ and $H^- = \{(x, y, z) | x^2 + y^2 - z^2 < 0\}$.

- For the region H^+ , we use $(x, y, z) = (r_+ \cosh u \cos v, r_+ \cosh u \sin v, r_+ \sinh u)$ (with $(r_+, u, v) \in (0, +\infty) \times \mathbf{R} \times [0, 2\pi)$) as the *hyperboloidal coordinate*;
- for H^- we use $(x, y, z) = (r_- \sinh u \cos v, r_- \sinh u \sin v, r_- \cosh u)$;
- for H^0 we use $(x, y, z) = (z \cos v, z \sin v, z)$ (with $(v, z) \in [0, 2\pi] \times \mathbf{R}$) to parameterize.

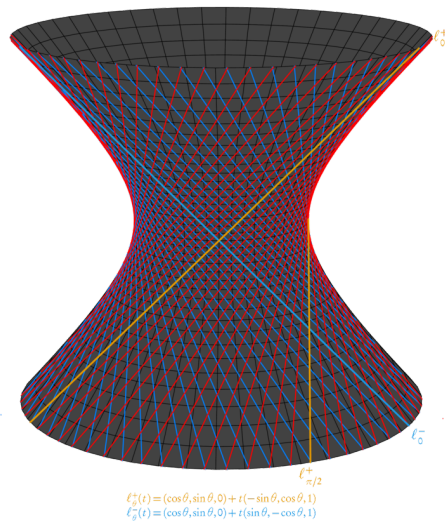


Figure 2: Quadric surfaces

Example 1.27. *

Use coordinate transformation to simplify the quadratic equation

$$45x^2 + 45y^2 - 8z^2 - 54xy + 36\sqrt{2}x - 108\sqrt{2}y + 32z + 184 = 0,$$

and identify the shape and position of this quadric surface.

1.6 Other common coordinate systems

There are several other coordinate systems (ways of parameterizing points in $\mathbf{R}^2$ or $\mathbf{R}^3$) which are commonly used, for instance *polar coordinate*, *cylindrical coordinate* and *spherical coordinate*. These are parallel to the Cartesian coordinate introduced above.

1.6.1 polar coordinate of $\mathbf{R}^2$

Pick a point O and a half line Ox (called the polar axis), for any P on the plane, let r denote $|OP|$ and $\theta \in [0, 2\pi)$ denote the angle from Ox to OP (by anticlockwise rotation), then (r, θ) is called the coordinate the polar coordinate of P .

If we take a special polar coordinate system (what we usually use): take O to be the origin of the plane, and the positive x axis (in the Cartesian coordinate) to be the polar axis, then the coordinate transformation between the two coordinate systems is

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \quad \text{and} \quad \begin{cases} r = \sqrt{x^2 + y^2} \\ \theta = \arctan \frac{y}{x} \end{cases}$$

Notice that $r \in [0, +\infty)$ and $\theta \in [0, 2\pi)$, and the transformation is one-to-one only on $(r, \theta) \in (0, +\infty) \times [0, 2\pi)$.

1.6.2 Cylindrical coordinate of $\mathbf{R}^3$

For $\mathbf{R}^3$, to parameterize a point P we can use the polar coordinate for its projection to Oxy plane and the height z . This is called the *cylindrical coordinate*:

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases} \quad \text{and} \quad \begin{cases} r = \sqrt{x^2 + y^2} \\ \theta = \arctan \frac{y}{x} \\ z = z \end{cases}$$

The triple $(r, \theta, z) \in [0, +\infty) \times [0, 2\pi) \times \mathbf{R}$ is called the *cylindrical coordinate system* of $\mathbf{R}^3$. Under this coordinate, $r = r_0$ represents a cylinder, and $\theta = \theta_0$ represents a half plane and $z = z_0$ represents a plane.

1.6.3 Spherical coordinate of $\mathbf{R}^3$

For $P \in \mathbf{R}^3$, let θ be the angle between OP with Oz , and P_1 be the projection of P to Oxy plane, then

$$\overrightarrow{OP} = \overrightarrow{OP_1} + |OP| \cos \theta \mathbf{k} = r \sin \theta \cos \varphi \mathbf{i} + r \sin \theta \sin \varphi \mathbf{j} + r \cos \theta \mathbf{k}.$$

The triple $(r, \theta, \varphi) \in [0, +\infty) \times [0, \pi] \times [0, 2\pi)$ is called the *spherical coordinate system* of $\mathbf{R}^3$. Under this coordinate, $r = r_0$ represents a sphere, $\theta = \theta_0$ represents a cone and $\varphi = \varphi_0$ represents a half plane. The usual seen formula for spherical coordinate is

$$\begin{cases} x = r \sin \theta \cos \varphi \\ y = r \sin \theta \sin \varphi \\ z = r \cos \theta \end{cases}, \quad \text{and} \quad \begin{cases} r = \sqrt{x^2 + y^2 + z^2} \\ \theta = \arccos \frac{z}{\sqrt{x^2 + y^2 + z^2}} \\ \varphi = \arctan \frac{y}{x} \end{cases}$$